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— *as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))*

— *as to the applicant's entitlement to claim the priority of the earlier application (Rule 4.17(iii))*

— *of inventorship (Rule 4.17(iv))*

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(54) **Title**: VISIBLY NON-INTRUSIVE DIGITAL WATERMARK BASED PROFICIENT, UNIQUE & ROBUST MANUAL SYSTEM FOR FORENSIC DETECTION OF THE POINT OF PIRACY (POP) OF A COPYRIGHTED, DIGITAL VIDEO CONTENT

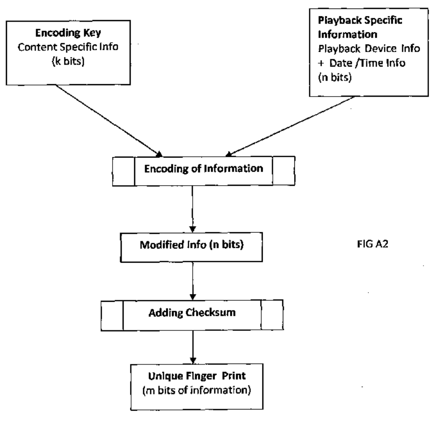

FIG A2

Fingerprint generation for Type II

(57) **Abstract**: The Invention involves embedding a unique digital fingerprint within the video such that the fingerprint can easily and uniquely identify the device which performed the playback and also the date and time of the playback and changes each time a different playback is selected for the same content and is positioned randomly being different in each frame of the video or other media.

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1.0 Introduction

As per media statistics, the Indian film industry suffers heavy loss to the tune of about Rs. 4000 crore per annum due to the piracy of the Indian movies.

The quantum of the box office collection of the first week is a yardstick to reveal the popularity of the movie. The huge collection figures of box office ill motivates pirates to cash on the popularity of the content by adopting unfair trade practices. Till date, globally various national governments have made major attempts to wipe out piracy in any form by enforcing strict Intellectual Property rights and laws. But, they have failed miserably due to the following factors:

1. *Lack of co-ordination between the multi nodal agencies involved.*
2. *Technical incompetence of the human resources handling the content and its subsequent piracy.*
3. *Vested interests of responsible and accountable persons.*
4. *Nexus between the pirates and exhibitors or manpower at various stages in the process of generation of the copyrighted print.*
5. *Access, availability, skill sets, privacy and time to handle the latest sophisticated hardware.*

The digital era has simplified the process of content creation, multiplication and delivery. Simultaneously it has enhanced the ease at which the buyer can illegally re-distribute the content, thus denying legitimate income to the content creator. Video and images published on the net are often victims of such piracy.

2.0 Piracy

Film industry being a money spinning industry, the demand for popular movies is ever increasing. Copyrighted content comes at a premium. Rising ticket rates, lack of free time at the disposal of viewer, limited choice of media are some of the challenges that affect the revenue of the exhibitors. Viewers at numerous video parlors spread over the entire nation wish to watch the latest and popular movies irrespective of quality of print, authenticity of the print, legal compliance of the print *etc.* This greed of the spectators to watch the first day first show at affordable cost indirectly drives the piracy market.

The revenue loss due to piracy has taken alarming shape not only due to the aforesaid factors but also due to the coming of new players in the media broadcasting domain *viz. set* top box manufacturers, operators, content uplinking/downloading organizations, tail end users (subscribers) at home/office *etc.* Specifically tail end home users have substantial time, privacy and knowledge to use equipments such as high definition handy cam to make unauthorized camera print for captive as well as commercial consumption.

The unethical methodology of accessing a multimedia content and its subsequent duplication, distribution for quick financial gains cannot be prevented by simple implementation of the strict copyright law.

Given the fact that pirates leverage the benefits of ultra modern IT tools at their disposal to defeat the copyright, this has motivated multimedia creators to develop technically innovative solutions based upon steganography messages, watermark or fingerprint to mark copyrighted content.

However, digital video watermarking technology cannot eliminate piracy totally. The fact that the logistics of movement of content through several nodes with ample time at disposal at each node, legal access to the secured content and availability of image capturing devices leads to piracy.

The invention described herein attempts to locate the precise time and geographical location of the point of piracy by analyzing a recording of the pirated video content, played back through a device implementing the watermarking technology. This is done with the aid of a unique software program to detect the watermark while the pirated content is played back.

3.0 Invention

There are many possible ways to distribute and play copyrighted content. For example, analog reels (to screen in a theatre), A DVD release (for home video) or broadcast on a TV network for a wider audience.

In spite of the prevailing piracy prevention or copy prevention techniques, not many have found wide spread deployment due to their prohibitive costs

The technique described here is simple to implement and very easy to adopt for wide spread deployment.

Besides, being robust this technique in combination with strict anti-piracy laws and stricter implementation can act as a huge deterrent to piracy.

The idea behind the invention is to use video content specific information along with the presentation details to create a unique digital fingerprint. This fingerprint is embedded as a visually non-intrusive watermark in the presented video signal. Any recordings of this video signal will contain the fingerprint. On recovery, this fingerprint uniquely identifies the source of the original presentation.

Watermarks can be categorized into three types: - non-blind watermark schemes, semi-blind watermarking schemes and blind watermarking schemes. If the original host image is required to reliably extract the embedded watermark, the scheme is non-blind. If the watermarking scheme needs the original watermark used before the watermark is embedded in the final video signal it is a semi-blind watermark. On the other hand, if the watermark is recoverable from the final video signal, without the need for the original image or any other information, it is a blind watermark.

The inventors' technique uses a semi-blind watermark, since the original video signal is required as a reference to locate the watermark in the pirated video signal.

The semi-blind watermark, in the inventors technique, which is embedded in the video signal uses following five key components.

a. *Content Identifier*
b. *Identifier of the Playback system used for presentation.*
c. *Date stamp (eight-digit format e.g. 30102008)*
d. *Time stamp (round the clock in hours e.g. 2300 hour)*

This information is presented on the screen in two different forms.

The first form (Type 1) consist of presenting the information (Items b,c and d) in a single shot, spread spatially, in the form of a grid. This grid is displayed in a visibly non-intrusive manner, at specific points during the video playback, the frequency of which is decided by a combination of the four above.

The second form (Type II) consists of encoding the information (Items b,c and d with knowledge of Item a) and displaying the individual bits over time, (spread temporally), in a visibly non-intrusive manner.

4.0 Methodology

The innovation aims at embedding a unique digital fingerprint within the video that is displayed on a large screen, projection surface or a TV screen, such that the fingerprint can easily and uniquely identify the device which performed the playback and also the date and time of the playback. The fingerprint changes each time a different playback is selected for the same content.

Figures A, B & C highlight the complete integrated flow chart based process of creation, embodiment and recovery of the fingerprint for detection of point of piracy.

Figure A1 and A2 contains the flowchart

4.1 Process to Create the Fingerprint

For Fingerprint Type I

(a) In order to create the fingerprint of Type I, 1st the device identification information, typically the "Device id" is selected.
(b) Also the "Date and Time" of when the playback started is selected (m bits of information)
(c) These m bits of information are then scrambled to form a unique m bit pattern.

For Fingerprint Type II

(a) In order to create the fingerprint of Type II, 1st the device identification information, typically the "Device id" is selected.
(b) Also the "Date and Time" of when the playback has started is selected (m bits of information).
(c) These "m bits" are then encoded with the (unique) content specific information "(k bits)".

(d) A "Checksum" is calculated for these m bits of information, The Checksum helps in ensuring that the recovered information is correct.

(e) The resulting "n bit" of encoded information is the fingerprint capable of identifying the device and instance of playback.

(f) The process of using the checksum and content specific information makes the fingerprint more robust.

FIG B contains the flowchart

4.2 Process of Using the Fingerprint

(a) Both these fingerprints are used to watermark the projected video signals.

(b) These watermarks are displayed at pseudo-random intervals during the length of playback.

(c) In case of both the fingerprint types, the points when they are displayed on screen are a function of Content Specific Information.

4.3 Act of Piracy

(a) Any recording of the video signals, projected using a device with the inventor's solution, will contain the fingerprints.

(b) With the availability of sophisticated duplication devices, such pirated content finds its way to the streets in matter of a couple of days of a theatrical release of a new film.

FIG C contains the flow chart

4.4 Process of Recovery of the Fingerprint

On acquiring a pirated copy of content, where the source of the pirated content is one of the devices with this fingerprinting technology enabled, the source information can be recovered in the following manner.

(a) Run the original content through a Decoder Program which displays the locations on the content where the fingerprint information can be found.

(b) Play the pirated content and look for the fingerprint information at those locations.

(c) Once the digital fingerprint is recovered (m bits in case of Type I and n bits in case of Type II), run it through the fingerprint decoder program.

(d) The original Encoding Key (Content Specific Info – k bits) used by the device to generate the fingerprint is used in the decoding process.

(e) The program checks for the integrity of the fingerprint and then decodes the device identification and date/time information.

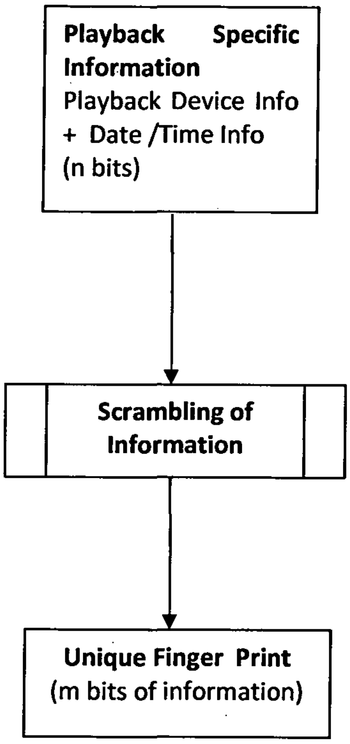
(f) The recovery process can make use of either Type I or Type II Fingerprint information, independently, for identification of the source.

(g) The further verification of the recovered information can be validated against the content playback logs of all devices available in the system.

We claim,

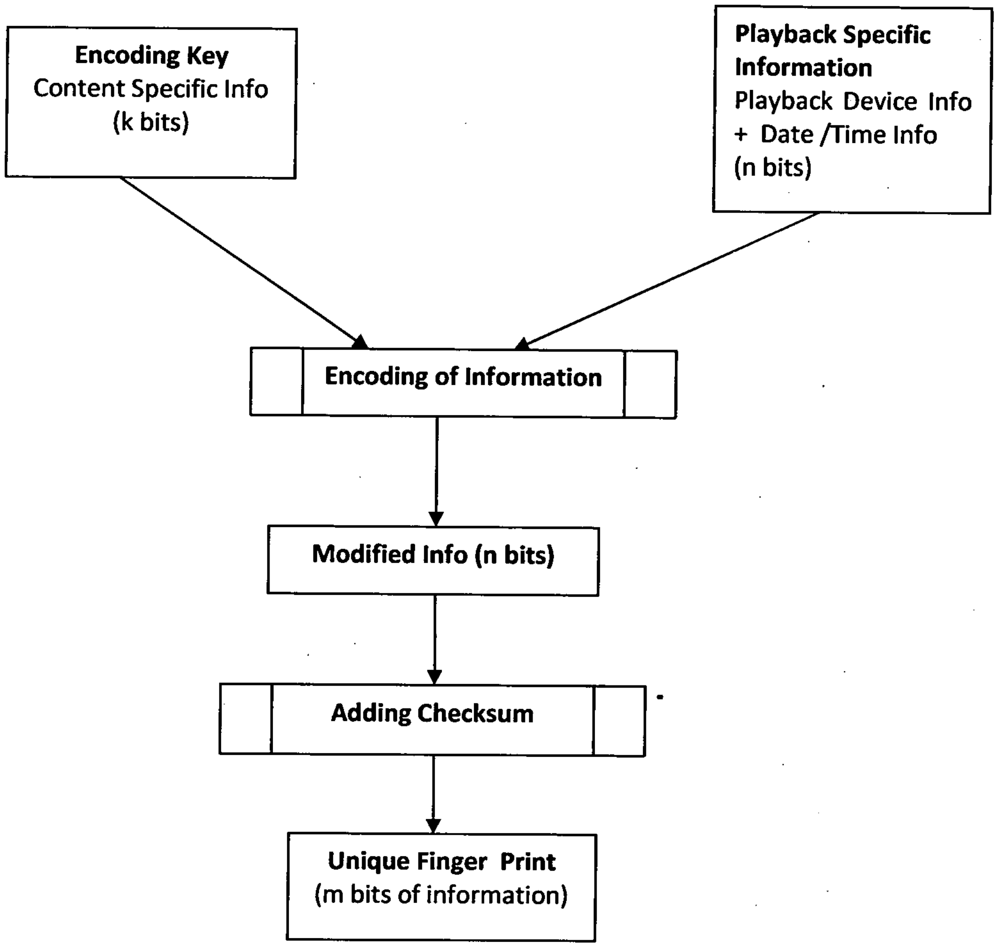
1. A method of embedding a unique digital finger print or water mark for forensic detection of the point of piracy (pop) of copyrighted digital video content comprising of
   a. the playback device information identifier –id
   b. date and time of presentation.
2. A method of embedding the unique fingerprint such that about 20 minutes of the pirated video is enough for successful recovery of source information.
3. Method as in claim 1 and 2 wherein the embedment of the digital fingerprint/watermark occurs pseudo randomly during the length of the video presentation.
4. Method as in claim 1 and 2 wherein the manner of embedment of the digital fingerprint / watermark can be picked up by a low resolution camera with resolutions as low as "CIF 352 X 288" .5 Method as in claim 1 and 2 wherein the watermark is a visually non-intrusive.
5. Method as in claim 1 and 2 wherein the embedded digital fingerprint/ watermark can be recovered even if the media has been re-encoded.
6. Method as in claim 1 and 2 above wherein the original content through a decoder program can display the locations on the content where the fingerprint information can be found.
7. Method as in claim 1 and 2 wherein once the digital fingerprint is recovered (n bits), and run through the fingerprint decoder program which checks for the integrity of the fingerprint and then decodes the device identification and date/time information.

Figure A1



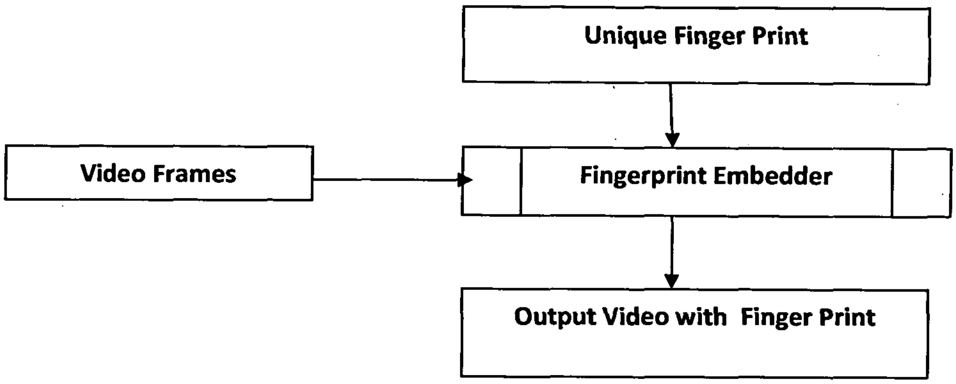
Fingerprint generation algorithm for Type I

FIG A2



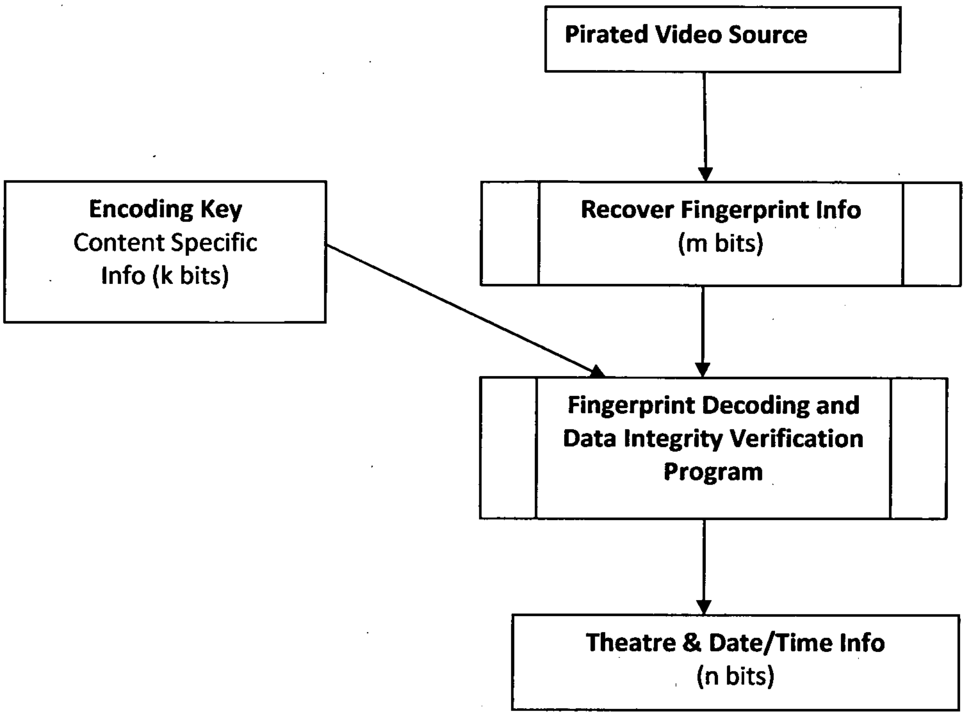
Fingerprint generation for Type II

FIG B



Embedding the fingerprint in the output video

FIG C



Recovery and Decoding of fingerprint